Problem 1. Show that additive inverses are *unique*. (That is, if $a \in \mathbb{R}$, then there is only one $b \in \mathbb{R}$ such that $a + b = 0$.)

Problem 2. Let $a, b, c \in \mathbb{R}$. If $a > b$ and $c > 0$, then $ac > bc$.

Problem 3. Either prove the following or provide a counterexample to show that it is false:
For all $a, b \in \mathbb{Z}$, if ab is even, then a is even or b is even.

Problem 4. Let $a, b \in \mathbb{Z}$. Assume that $a \equiv 7 \pmod 8$ and $b \equiv 3 \pmod 8$.

1. Prove that $a + b \equiv 2 \pmod 8$.
2. Prove that $ab \equiv 5 \pmod 8$.
3. Find an integer r with $0 \leqslant r < 8$ such that $a^{14} \equiv r \pmod 8$.

Problem 5. Prove the following theorem:
Let $n \in \mathbb{Z}$. If $n^2 - 1$ is divisible by 2, then $n^2 - 1$ is divisible by 4.

Problem 6. Recall that a real number x is *rational* if we can write $x = \frac{p}{q}$ for some $p, q \in \mathbb{Z}$ with $q \neq 0$. We say that x is *irrational* if it is impossible to write x in this way. Either prove the following statement or provide a counterexample to show that it is false:
Let $x \in \mathbb{R}$ with $x \geqslant 0$. If x is irrational, then $\sqrt{x}$ is irrational.

Problem 7. Prove or provide a counterexample to the following two statements:

1. The sum of two rational numbers is rational.
2. The sum of two irrational numbers is irrational.

Problem 8. Either prove the following or provide a counterexample to show that it is false:
Let $n \in \mathbb{Z}$. If n is odd, then $n^2 \equiv 1 \pmod 8$.

Problem 9. Complete Exercise 20 on pages 129 and 130 of the textbook (*Mathematical Reasoning* by Sundstrom).

Problem 10. Do not turn these in, but I recommend doing the progress checks for sections 3.3 and 3.4 in the textbook.